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Prenatal stress, excitatory-inhibitory imbalance, and ADHD risk: a hypothesis-driven perspective on psilocybin-induced neuroplasticity

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Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental condition in which prenatal stress and long-lasting disruptions of excitatory–inhibitory (E/I) balance have been implicated as key vulnerability factors. Although established pharmacological and behavioral treatments are effective for many individuals, they are not universally successful and do not directly target upstream neurodevelopmental mechanisms. In this hypothesis-driven perspective, we examine whether psilocybin-induced neuroplasticity could theoretically modulate stress-related neurodevelopmental risk pathways relevant to ADHD. Rather than presenting psilocybin as an evidence-based intervention, we synthesize findings from related preclinical and clinical literatures to explore conceptual plausibility. Preclinical studies in non-ADHD models indicate that psilocybin can induce rapid and sustained synaptic plasticity, alter cortical E/I dynamics, and reverse stress-associated structural and functional alterations. Human clinical trials in adults—primarily in mood, trauma-related, and substance use disorders—demonstrate durable changes in emotional regulation, cognitive flexibility, and large-scale brain network organization, processes that overlap with neural systems implicated in ADHD. Research into the use of psilocybin for ADHD is in its early stages, with emerging, largely self-reported, and preliminary studies suggesting potential benefits for managing symptoms like inattention, impulsivity, and emotional dysregulation. We therefore frame psilocybin as a speculative (secondary/tertiary) approach that could, in principle, be explored to probe mechanisms of E/I rebalancing and neuroplasticity. Key mechanistic uncertainties—including the state-dependent effects of psilocybin on excitation and inhibition and the possibility of exacerbating existing imbalances—are explicitly discussed. Ethical and developmental considerations, particularly regarding vulnerable populations, are emphasized as critical constraints on translation. Finally, we propose a translational research roadmap encompassing preclinical prenatal-stress models, biomarker-driven pilot studies in ADHD, and multimodal outcome measures integrating neuroimaging, electrophysiology, and molecular indices. By clearly distinguishing established evidence from hypothesis, this perspective aims to stimulate rigorous and ethically grounded research rather than to advocate premature clinical application.

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INTRODUCTION

Health is best viewed as a dynamic and multidimensional capacity, rather than merely the absence of disease [1]. It encompasses more than just the lack of illness; It is a multidimensional capacity that enables children and adults to navigate everyday challenges, fend off infections, cope with adversity, experience personal well-being, and engage with their environment in ways that foster positive development [2]. Experiences during early life play a pivotal role in shaping children's neural, behavioral, and psychological development, with lasting impacts across multiple domains of life [1, 3]. Secure, stable, and supportive environments in the early life period are vital for fostering long-term health and well-being [4]. These settings—characterized by consistent, responsive relationships and nurturing physical and emotional conditions—facilitate healthy brain development and effective stress regulation, thereby enhancing resilience and reducing the risk of chronic diseases in adulthood during this sensitive period [5, 6]. In contrast, adverse childhood experiences—such as abuse, neglect, or family dysfunction—can disrupt the formation of brain structures and biological systems [7, 8]. This disruption may lead to toxic stress, which has been linked to increased risk for a variety of health conditions, including heart disease, diabetes, depression, and substance misuse later in life [9]. Understanding the beginnings of adult illnesses and tackling them early in life is essential for shifting our health care system from a focus on treating sickness to one that promotes overall wellness [10, 11]. But how early can these effects begin? Could the roots of long-term health outcomes be traced even further back—before birth? In response to this question, the British physiologist Barker was the first to propose the fetal programming hypothesis [12].

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"Programming" is understood as the process through which a stimulus or disruption occurring during critical periods of development, when biological systems are particularly plastic and sensitive to environmental influences, can lead to long-lasting or even permanent effects. Notably, this notion conveys the idea that the physiological adaptation by which prenatal exposure to nutritional and environmental influences shapes fetal development, resulting in metabolic alterations and heightened risk for chronic conditions later in life [13, 14]. In this concept, adaptation refers to the physiological and developmental adjustments the fetus makes in response to prenatal stressors. In the short term, such adaptations may include down-regulation of endocrine, metabolic, or organ functions in order to reduce growth rate and minimize nutrient consumption [15, 16]. In the long term, these changes can influence gene expression, as well as cellular differentiation and proliferation, ultimately leading to structural and functional alterations in specific tissues and vital organs—many of which may be lasting or even permanent [17, 18]. Accordingly, fetal programming is often conceptualized as a form of predictive adaptive response, through which the fetus adjusts development based on anticipated postnatal environments [19, 20]. Prenatal exposure to elevated maternal stress hormones, particularly cortisol, represents a well-documented example of predictive adaptive response. When a fetus develops in a high-stress intrauterine environment, its hypothalamic-pituitary-adrenal (HPA) axis may become programmed to be more reactive to stress, through increased expression of glucocorticoid receptors in brain regions involved in emotional regulation [21, 22]. While this adaptation may enhance survival in similarly adverse postnatal conditions, it may also predispose the individual to emotional dysregulation, heightened anxiety, or depressive symptoms in later life, particularly when the early predictions do not match the actual postnatal environment [23]. Building on this understanding, the Eco-Bio-Developmental model offers a comprehensive framework for explaining how early environmental experiences become biologically embedded, influencing long-term health outcomes across the lifespan [24]. This model integrates three interrelated domains—ecological, biological, and developmental—to illustrate how early-life conditions, whether adverse or supportive, interact with genetic and physiological systems to shape developmental trajectories and future vulnerability to disease [25]. The ecological dimension encompasses the child's external environment, including caregiver relationships, family dynamics, socioeconomic status, and exposure to stress or supportive interactions [26]. The biological dimension involves the physiological systems that respond to these environmental signals—particularly the neuroendocrine, immune, and metabolic systems—through mechanisms such as epigenetic modification, HPA axis programming, and inflammatory responses [20, 27]. The developmental dimension underscores the significance of sensitive periods in early life, during which biological systems exhibit heightened plasticity and are particularly vulnerable to environmental input [28–30]. Over time, these changes can become biologically embedded, increasing the risk of emotional dysregulation, impaired executive functioning, and chronic physical or mental health conditions in the absence of buffering protective relationships [30, 31]. Against this background, prenatal stress has emerged as a significant risk factor for Attention-Deficit/Hyperactivity Disorder (ADHD), a prevalent neurodevelopmental condition characterized by deficits in attention, impulse control, and emotional regulation [32]. Converging evidence implicates disruptions in excitatory–inhibitory (E/I) balance within frontostriatal and corticolimbic circuits as a key neurobiological mechanism linking early stress exposure to ADHD-related phenotypes [33, 34]. The present article adopts a hypothesis-driven, perspective-based approach rather than a comprehensive evidence-based review. Its primary aim is to synthesize current knowledge on prenatal stress, E/I imbalance, and ADHD risk, and to explore—at a conceptual level—the plausibility that interventions targeting neuroplasticity could modulate stress-related developmental trajectories.

Within this context, we discuss psilocybin-induced neuroplasticity as a conceptual and mechanistic framework for future investigation, rather than as an established or recommended clinical intervention. Importantly, rigorous, placebo-controlled clinical trials of psilocybin for ADHD are largely absent, and the existing literature is primarily limited to naturalistic, user-driven reports rather than controlled experimental studies. Moreover, no studies have directly examined psilocybin in prenatal stress models with ADHD-relevant behavioral, cognitive, or circuit-level outcomes—a limitation that is explicitly acknowledged throughout this manuscript. Accordingly, the present perspective does not address primary prevention strategies aimed at reducing risk prior to disease emergence. Instead, it situates psilocybin within secondary and tertiary prevention frameworks, wherein interventions are applied after vulnerability, early symptoms, or established dysfunction have emerged, with the goal of modulating maladaptive plasticity, attentional control networks, and executive-function circuitry. While this framework draws on broader literature linking early-life stress to long-lasting neurodevelopmental alterations, the present focus is specifically on psychological gestational stress—such as maternal anxiety, depression, and perceived stress during pregnancy—and its downstream implications for attentional and executive networks relevant to ADHD. This emphasis reflects both the high prevalence and empirical tractability of prenatal psychological stress, as well as growing recognition of its enduring neurobiological impact. By clearly distinguishing established findings from hypothesis-driven extrapolation, this article aims to delineate critical knowledge gaps, ethical boundaries, and experimental trajectories that may guide future research, rather than to advocate premature clinical application.

CONCEPTUALIZING PRENATAL STRESS

Stress is a multifaceted biopsychosocial phenomenon that arises when an individual perceives a situation as threatening or overwhelming and believes that their available coping resources are insufficient to manage it. While acute stress may activate adaptive behavioral and physiological systems that support survival and effective functioning, chronic or intense stress can result in prolonged activation of these systems—particularly the HPA axis and immune pathways—leading to dysregulation and increased vulnerability to physical and mental health disorders over time [35, 36]. Importantly, early and repeated exposure to stress, especially during sensitive developmental windows, can become biologically embedded, altering gene expression, brain architecture, and long-term stress responsivity [27]. To fully understand the nature of stress, it is essential to differentiate between stressors—external or internal stimuli that have the potential to disrupt homeostasis—and the subjective perception of stress, which determines the intensity and pattern of the physiological and psychological responses [37]. Stressors may take various forms, including acute stressors (e.g., injury, a job interview), chronic stressors (e.g., persistent financial hardship), traumatic stressors (e.g., exposure to violence or natural disasters), and daily hassles (e.g., workload or minor frustrations) [38, 39]. In addition to categorizing stressors by their duration and nature, it is essential to distinguish between stressors—the external or internal events that pose a challenge to homeostasis—and the stress responses that such stimuli may provoke. Stressors may be classified as acute, chronic, traumatic, or daily hassles, and they vary widely in severity, frequency, and subjective meaning [40, 41]. Importantly, how stress is conceptualized theoretically also shapes our understanding of its impact on health and development [42]. Among the various forms of stress encountered across the lifespan, prenatal stress has attracted growing scientific interest

due to its occurrence during a critical window of development and its long-lasting effects on offspring health [23, 43–46]. Reported prevalence rates of prenatal stress vary widely across studies, ranging from 5.5 to 78%, depending on sociocultural and economic factors [47]. Defined as maternal exposure to physical, psychological, or environmental stressors during gestation, prenatal stress can influence fetal development through alterations in the intrauterine environment [48]. Even in the absence of overt pathology, persistent maternal stress has been linked to disruptions in the regulation of stress-related systems in the fetus, including the HPA axis and immune function [49]. These early disruptions may increase vulnerability to adverse outcomes such as emotional dysregulation, cognitive impairments, and neurodevelopmental disorders [48]. While the exact mechanisms remain under investigation, emerging evidence suggests that the effects of prenatal stress may be biologically embedded, with implications that extend into childhood and even adulthood [50].

From prenatal stress to ADHD

Building on the evidence surrounding prenatal influences, several studies now focus on the association between maternal stress during gestation and the onset of ADHD-related behaviors in offspring [51–54]. "ADHD is a neurodevelopmental condition with a strong genetic predisposition, characterized by persistent patterns of inattention, disorganization across multiple domains, and symptoms of hyperactivity and impulsivity" [55]. According to a recent systematic review, the global prevalence of ADHD is estimated at 7.6% in children aged 3 to 12 years and 5.6% in adolescents aged 12 to 18 years [56]. Prospective studies have shown that 30–60% of symptoms and functional impairments associated with ADHD persist into adulthood, which has led to the growing recognition of adult ADHD as a distinct clinical entity [57, 58]. The prevalence of ADHD among adults is estimated at 2–4%, yet despite its high frequency, it remains significantly underdiagnosed and undertreated [59, 60]. Clinically, adult ADHD manifests similarly to its childhood presentation, including the classic triad of inattention, hyperactivity, and impulsivity. However, symptoms of hyperactivity and impulsivity tend to externalize, while inattentiveness may be masked by comorbid anxiety or compensatory mechanisms resembling OCD-like behaviors. Substance use is highly prevalent among individuals with ADHD, often as a result of emotion regulation deficits and impulsivity, or as a form of self-medication [61]. Accordingly, the elevated risk of developing ADHD following exposure to prenatal stress represents a critical and noteworthy concern in the context of neurodevelopmental vulnerability. Within this clinical context, prenatal stress is increasingly recognized as a developmental risk factor that may interact with genetic susceptibility to influence ADHD-related outcomes, rather than as a deterministic cause. Exposure to maternal psychological stress during gestation has been associated with alterations in offspring stress responsivity, executive functioning, and attentional control—domains that overlap with core ADHD symptom dimensions [51–53]. These findings support a vulnerability-based framework in which prenatal stress contributes to neurodevelopmental risk by shaping neural circuits involved in cognitive and emotional regulation. Given the enduring influence of prenatal stress on neurodevelopmental trajectories, there is growing interest in identifying interventions that target downstream mechanisms such as neural plasticity and circuit flexibility. Psilocybin, a serotonergic psychedelic compound, has been shown in preclinical models to induce synaptogenesis, dendritic remodeling, and upregulation of brain-derived neurotrophic factor (BDNF), processes associated with experience-dependent plasticity [62, 63]. Importantly, these effects have been demonstrated in stress-related or affective models rather than in ADHD-specific or prenatal stress paradigms. Accordingly, any discussion of psilocybin in the context of ADHD must remain speculative. Rather than positioning psilocybin as a therapeutic or preventive intervention, its relevance here lies in its utility as a *theoretical probe* for understanding how pharmacologically induced plasticity might interact with stress-altered neural systems. Whether such plasticity would normalize, have no effect on, or potentially exacerbate ADHD-relevant circuit dysfunction remains unknown. This uncertainty underscores the need for rigorous preclinical testing in prenatal stress models and cautious, biomarker-driven investigation in adult ADHD populations before any clinical implications can be considered.

EXCITATORY–INHIBITORY (E/I) IMBALANCE AS A MECHANISTIC PATHWAY

One of the central neurobiological mechanisms linking prenatal stress to later risk of ADHD is the disruption of the delicate balance between E/I signaling in the developing brain. The regulation of excitatory glutamatergic neurons and inhibitory γ-aminobutyric acid (GABA) interneurons is fundamental for cortical maturation, synaptic pruning, and the establishment of functional neural circuits [33, 34, 64]. Even subtle alterations in this balance during sensitive developmental periods may result in atypical cortical oscillations, impaired executive function, and behavioral phenotypes consistent with ADHD [65]. During early brain development, GABA initially exerts a depolarizing, excitatory action on immature neurons due to the predominance of the Na^+-K^+-Cl^- cotransporter 1 (NKCC1), which maintains elevated intracellular chloride levels. As the brain matures, a developmental 'switch' occurs with the upregulation of the K^+-Cl^- cotransporter 2 (KCC2), leading to chloride extrusion and the establishment of GABA's classical hyperpolarizing, inhibitory function [66, 67]. This transition is essential for the maturation of neural circuits, the refinement of synaptic connections, and the emergence of synchronized network oscillations that underlie higher cognitive functions [68]. Disruptions in the timing or mechanisms of the GABAergic switch have been implicated in neurodevelopmental disorders such as autism, epilepsy, and ADHD [69, 70]. Importantly, interventions that restore the excitatory-to-inhibitory GABA transition, such as NKCC1 antagonists, have shown promise in correcting abnormal network activity and associated behavioral phenotypes [71]. Consistent with the critical role of the GABAergic E/I switch in neurodevelopment, recent studies using a 22q11.2 microdeletion syndrome (22q11.2 DS) mouse model (Lgdel +/−) have revealed embryonic disruptions in this process [71]. Individuals with 22q11.2 DS exhibit cognitive and behavioral impairments, developmental delays, and an increased risk for neuropsychiatric disorders. In neurons derived from Lgdel +/− embryos, premature alterations in the expression of the chloride cotransporters NKCC1 and KCC2 were observed, indicating a delayed GABA polarity switch. These molecular changes were accompanied by aberrant neurite outgrowth, loss of synchronized spontaneous network activity, and impaired homeostatic plasticity, as demonstrated by large-scale spiking recordings. Parallel studies in Dgcr8 +/− cultures suggest that the dgcr8 gene contributes partially, though not exclusively, to these phenotypes [72]. Importantly, treatment with bumetanide, an NKCC1 inhibitor, reduced the hyperexcitability of GABAA receptor signaling and restored homeostatic network plasticity [72]. Overall, these findings implicate a delayed embryonic GABA switch as a potential mechanistic driver of early neurodevelopmental deficits in 22q11.2 DS, highlighting the GABA polarity transition as a prospective target for therapeutic intervention [73]. Together, these findings support the view that disruption of the developmental GABA switch can function as a mechanistic vulnerability factor, predisposing neural circuits to long-lasting alterations in excitability, synchrony, and adaptive capacity [73]. However, it is important to emphasize that E/I imbalance does not imply a single directional abnormality; depending on developmental stage, brain region, and stress context, prenatal adversity may produce

hyperexcitable, hypoexcitable, or dysregulated oscillatory states. Early-life stress, including prenatal stress, has been shown to interfere with the maturation of inhibitory circuits and the timing of the excitatory-to-inhibitory GABA transition, thereby increasing susceptibility to neurodevelopmental conditions such as ADHD—not as a deterministic outcome, but as one pathway among multiple interacting genetic and environmental influences.

Prenatal stress and E/I circuit disruption

Prenatal stress exerts profound effects on the HPA axis, leading to increased maternal cortisol levels that cross the placenta and alter fetal brain development [23]. Excessive or prolonged glucocorticoid exposure during gestation has been associated with alterations in neuronal proliferation, differentiation, and migration, including effects on GABAergic interneuron development [74]. In particular, changes in the maturation and distribution of parvalbumin-positive interneurons—key regulators of cortical excitability, timing, and network synchrony—have been reported in several preclinical models of prenatal stress [75, 76]. Parallel evidence indicates that prenatal stress may upregulate glutamatergic signaling, through increased expression of NMDA receptor subunits (e.g., GRIN1, GRIN2B) and changes in glutamate transporter function, further tipping the E/I balance toward hyperexcitability [77]. Rather than uniformly increasing excitation, these molecular changes are thought to reflect dysregulated excitatory signaling that may vary by brain region, developmental stage, and stress intensity. Such alterations have the potential to perturb E/I coordination during critical periods of circuit formation.

Animal studies provide converging support: rodent offspring exposed to gestational stress show reduced GABAergic tone in the prefrontal cortex and hippocampus, accompanied by enhanced glutamatergic transmission and impaired inhibitory control [78]. These neurochemical alterations align with behavioral outcomes including heightened impulsivity, increased locomotor activity, and attention deficits—hallmarks of ADHD-like phenotypes [79]. Importantly, these phenotypes should be interpreted as analogues rather than direct equivalents of human ADHD symptoms. Experimental reviews further highlight that prenatal stress can promote network-level dysregulation through combined effects on GABAergic and glutamatergic signaling, potentially mediated by neuroinflammatory processes [80]. A rat-based study recently demonstrated changes in gene expression associated with excitatory and inhibitory neurotransmission in the prefrontal cortex and amygdala following maternal stress, implicating these molecular shifts in emotional dysregulation [78]. Such findings support a model in which prenatal stress alters the developmental calibration of neural circuits, rather than producing a fixed excitatory or inhibitory pathology. At the systems level, mesoscale imaging studies using voltage-sensitive dyes have shown that prenatal stress can reduce cortical sensory responsiveness and disrupt resting-state functional connectivity. Specifically, stressed offspring exhibit weakened inter-modular coordination, enhanced intra-modular coupling, and reduced global network efficiency. These alterations suggest impaired integration and flexibility of large-scale cortical networks, consistent with disrupted E/I dynamics at the circuit level. Collectively, these changes may increase vulnerability to stress-related neuropsychiatric outcomes in adulthood [81] and raise the likelihood of ADHD-like phenotypes in preclinical models, without implying a deterministic developmental trajectory.

ADHD and E/I imbalance in human studies

Human neuroimaging studies largely corroborate preclinical evidence by demonstrating altered E/I dynamics in individuals with ADHD. Proton magnetic resonance spectroscopy (^{1}H-MRS) studies have reported reduced cortical GABA concentrations alongside elevated glutamate or glutamine levels in key regions involved in cognitive control and reward processing, including the anterior cingulate cortex, prefrontal cortex, and striatum, in both children and adults with ADHD [82]. However, regional specificity appears to be critical. In a recent high-field (7T) MRS study of unmedicated children with ADHD, reduced GABA/creatine ratios were specifically observed in the striatum, whereas no significant group differences were detected in the anterior cingulate, dorsolateral prefrontal, or premotor cortices. Glutamate concentrations did not differ between groups, and no associations with behavioral symptoms were found. These findings suggest that reduced striatal GABA may represent a potential neurochemical marker of impaired inhibitory control in ADHD and could inform the development of targeted GABA-related interventions [83].Converging evidence from functional MRI and EEG studies further supports the presence of E/I dysregulation in ADHD by revealing atypical gamma-band oscillations, which critically depend on the synchronized activity of GABAergic interneurons and are essential for attentional control [84]. Extending these observations beyond traditional task-based paradigms, recent EEG work using naturalistic movie stimuli demonstrated that children with ADHD not only exhibit reduced inter-subject correlations in gamma-band oscillations during social perception tasks but also show altered spatial activation patterns. Specifically, typically developing controls showed stronger right parietal activation, whereas children with ADHD displayed weaker responses in the left occipital lobe. Together, these findings indicate that disrupted gamma-band synchrony contributes not only to attentional impairments but also to deficits in social information processing, highlighting gamma oscillations as promising neural markers for diagnosis and treatment monitoring [85]. Importantly, E/I imbalance is not unique to ADHD but appears to represent a shared neurobiological mechanism across neurodevelopmental disorders. Accumulating evidence suggests that disorders such as ADHD, ASD, and related conditions show substantial overlap at the genetic and circuit levels. The risk of these disorders is increased by copy number variant (CNV) syndromes—phenotypically heterogeneous genetic alterations present in a small proportion of the population. Current neurodevelopmental models propose that disruption of E/I balance may be a core etiological mechanism underlying these disorders and may be further exacerbated in individuals carrying CNVs [86]. Despite the clinical efficacy of current treatments targeting catecholaminergic transmission, a substantial subset of patients remains only partially responsive, often necessitating prolonged trial-and-error treatment strategies. Recent preclinical and clinical studies suggest that early-life E/I imbalance within fronto-striatal circuits may drive persistent neuroanatomical and functional alterations, contributing to symptom continuity into adulthood. This framework has prompted proposals for precision medicine approaches that incorporate E/I balance as a biomarker, emphasize the development of GABA-modulating therapeutics, and utilize neuroimaging and electrophysiological tools to track treatment effects across the lifespan [87]. Notably, these alterations appear to mediate core ADHD symptoms: reduced inhibitory tone has been linked with impulsivity and hyperactivity, while excessive glutamatergic drive contributes to distractibility and deficits in executive functioning [65, 88].Taken together, evidence from both animal models and human studies converges on E/I imbalance as a central mechanistic link between early neurodevelopmental perturbations—including prenatal stress exposure—and the emergence of ADHD-related behavioral phenotypes. In parallel, several recent studies have begun to explore strategies aimed at preventing or restoring E/I balance in the human brain. 1**. Transcranial Direct Current Stimulation (tDCS) and Transcranial Electrical Stimulation (tES):** Clinical studies indicate that tDCS, a non-invasive brain stimulation method, can modulate E/I activity and restore balance in humans, particularly in neuropsychiatric and neurodegenerative conditions such as ASD, ADHD,

epilepsy, and food or opiate addiction [89–91]. Preclinical research further clarifies the mechanisms of various tES modalities. Animal studies further elucidate the mechanisms of tES variants like tDCS, transcranial alternating current stimulation (tACS), and transcranial random noise stimulation (tRNS. Each operates differently—tDCS influences polarity-dependent plasticity, tACS can entrain frequency-specific oscillations, and tRNS modulates GABAergic activity [92]. Although much of the mechanistic insight derives from animal models, these findings underscore the translational potential of tES-based interventions. **2. Ketamine in Treatment-Resistant Depression**: Evidence from patients with treatment-resistant major depressive disorder reveals that subanesthetic ketamine (a NMDA receptor antagonist) infusion influences cortical E/I dynamics. Advanced computational modeling revealed that ketamine shifts neural activity toward a more stable balance of E/I, and these changes were closely associated with improvements in depressive symptoms [93]. These observations support the broader concept that restoring E/I equilibrium may represent a viable treatment strategy across multiple psychiatric conditions. **3. Finally, novel biomarker-driven approaches** are emerging to non-invasively estimate E/I balance in humans. Recent work combining in silico modeling with fMRI-derived metrics identified the Hurst exponent (H) as a proxy for synaptic E/I dynamics. Using this approach, autistic males—but not females—exhibited increased excitation in the medial prefrontal cortex. Such biomarker strategies may provide powerful tools for early risk stratification, longitudinal monitoring, and evaluation of E/I–targeted interventions in future clinical trials [94].

DEVELOPMENTAL TIMING AND SENSITIVE WINDOWS

The impact of prenatal stress on E/I regulation is particularly pronounced because gestation represents a critical window for interneuron migration, synaptogenesis, and network maturation [95, 96]. Disruptions during this window can lead to long-lasting structural and functional consequences, including altered cortical thickness, atypical connectivity, and reduced synaptic plasticity. Importantly, these changes do not simply resolve with postnatal development; instead, they become biologically embedded, shaping vulnerability to neurodevelopmental disorders including ADHD, autism spectrum disorder, and mood disorders [97]. Evidence from animal models underscores the importance of precise developmental timing. In rodents, the second and third weeks of gestation represent highly sensitive periods for the neurobiological effects of prenatal stress. A study by Sadaghiani and Saboory demonstrated that prenatal stress potentiated pilocarpine-induced epileptic behaviors in infant rats in a time- and sex-dependent manner [98]. Prenatal stress was applied during the first, second, or third week of gestation, as well as throughout the entire pregnancy. Stress exposure during mid- and late gestation produced the most pronounced effects, including shorter latency to seizure onset, increased frequency of tonic–clonic seizures, and higher mortality rates in 19-day-old pups. All stress paradigms elevated offspring corticosterone levels —by up to eight-fold—indicating sustained activation of the hypothalamic–pituitary–adrenal (HPA) axis. Notably, female offspring exhibited greater resilience, likely reflecting lower HPA axis sensitivity and sex-specific stress buffering mechanisms [98]. A longitudinal neuroimaging study from the European Longitudinal Study of Pregnancy and Childhood cohort found that prenatal stress during the first half of pregnancy was associated with reduced volumes of the basal and cortical nuclei of the amygdala in young adult offspring, independent of sex, maternal, and perinatal factors. Notably, amygdala nucleus volume showed a quadratic relationship with depressive symptoms in adulthood, with both smaller and larger volumes linked to higher Beck Depression Inventory scores. These findings suggest that early gestational stress has lasting effects on amygdala development, potentially increasing vulnerability to depression later in life [99]. The gestational timing of stress exposure therefore emerges as a critical determinant of neurodevelopmental outcome. Although fetal brain development has been extensively studied, relatively few investigations have examined how the gestational period of stress exposure influences offspring outcomes. Evidence suggests that disturbances occurring early in pregnancy may result in more severe neurological consequences than later stressors [100], potentially through disruptions in placental signaling, glucocorticoid regulation, and early neural patterning [101]. Supporting this view, Davis and Sandman reported that elevated maternal cortisol levels during early gestation predicted poorer cognitive performance in offspring, whereas higher cortisol exposure during late gestation was paradoxically associated with improved cognitive outcomes [102]. Interestingly, an opposite pattern has been observed for emotional and behavioral problems, with early gestational stress conferring greater risk [100]. In parallel, studies examining antenatal corticosteroid administration—most commonly administered during the late second to early third trimester —have reported an increased risk of mental and behavioral disorders in offspring, even in sibling-comparison designs, underscoring the sensitivity of late gestational brain development to glucocorticoid exposure [103]. Importantly, prenatal stress does not deterministically lead to adverse outcomes. Rather, it appears to increase offspring sensitivity to postnatal environmental conditions, functioning as a form of developmental programming. Consequently, when prenatal stress is followed by nurturing, supportive postnatal environments, it may even facilitate adaptive or resilient developmental trajectories [100].This perspective aligns with differential susceptibility and predictive adaptive response models, emphasizing that early-life stress interacts dynamically with later environmental inputs to shape neurodevelopmental risk or resilience.

PSILOCYBIN AND NEUROPLASTICITY: A THERAPEUTIC WINDOW

Evidence suggests that psilocybin, a naturally occurring tryptamine compound found in *Psilocybe* mushrooms, may be effective in the treatment of depression, anxiety, and addiction, with clinical benefits persisting for months or even years following a single or limited number of administrations. These sustained effects distinguish psilocybin from conventional pharmacotherapies and are hypothesized to arise from its ability to rapidly induce durable neuroplastic changes rather than continuous receptor occupancy [104]. Accordingly, psilocybin has re-emerged as a compound of intense scientific interest due to its unique capacity to promote rapid and persistent alterations in mood, cognition, and large-scale neural organization [105, 106]. Pharmacologically, psilocybin is rapidly metabolized into psilocin, which acts as a partial agonist at serotonin 5-HT2A receptors, with additional affinity for 5-HT1A, 5-HT2C, and other serotonergic sites [107, 108]. Through this receptor profile, psilocybin profoundly alters cortical network activity, enhances synaptic remodeling, and modulates emotional processing (Fig. 1) [109].

Psilocybin and structural and functional plasticity

Preclinical studies demonstrate that psychedelics, including psilocybin, promote structural plasticity by increasing dendritic spine density, enhancing synaptogenesis, and stimulating neurite outgrowth [110]. These effects are partly mediated by BDNF and downstream signaling pathways such as mTOR and ERK, which regulate cytoskeletal remodeling and synaptic stability [108, 111]. At the systems level, psilocybin reorganizes large-scale brain networks, most notably by reducing hyperconnectivity within the default mode network (DMN) and enhancing global connectivity across cortical and subcortical regions [112, 113]. Such network-

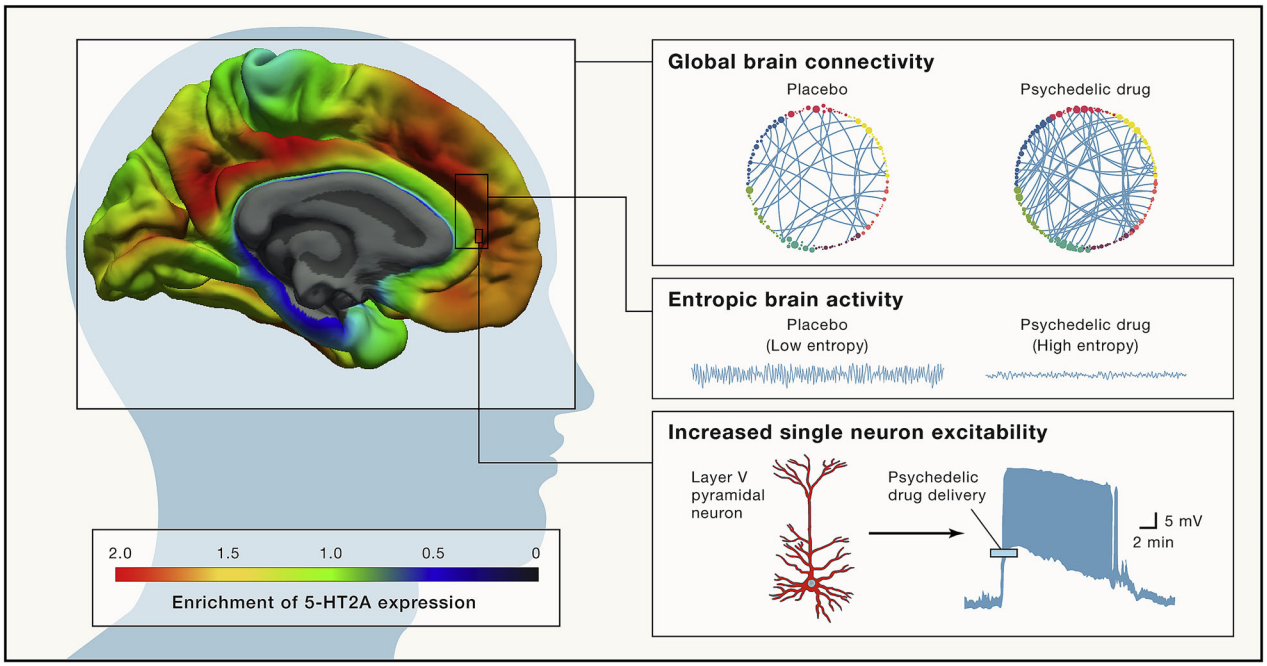


Fig. 1 Three Levels of Psilocybin's Action: Psilocybin, like other serotonergic psychedelics, primarily activates 5-HT2A receptors located on cortical layer 5 pyramidal neurons. This activation induces strong depolarization and rapid repetitive firing of these cells (lower inset). Since layer 5 pyramidal neurons coordinate cross-cortical communication, their heightened activity profoundly alters cortical information processing. Electrophysiological recordings, including magnetoencephalography and electroencephalography, show that psilocybin disrupts normal rhythmic patterns, producing a state of desynchronization or increased cortical entropy (middle inset). In addition, layer 5 neurons are central to generating "top-down" perceptual and cognitive predictions (priors) that typically constrain brain function. Under psilocybin, these priors are relaxed, enabling the brain to move beyond rigid and predictable processing modes. This shift results in greater global connectivity (top inset), supporting novel associations between memories, emotions, behaviors, and beliefs. Such reorganization of brain networks is thought to underlie the therapeutic effects of psilocybin in psychiatric conditions such as depression and addiction, where maladaptive patterns of neural activity dominate [108]; https://doi.org/10.1016/j.cell.2020.03.020.

level reconfiguration is hypothesized to support cognitive flexibility, emotional recalibration, and the enduring clinical benefits observed following psilocybin administration.

In a clinical trial by Carhart-Harris et al. psilocybin was studied using task-free fMRI to explore the transition from normal consciousness to a psychedelic state. The study involved 15 healthy volunteers scanned with arterial spin labeling (ASL) and another 15 with Blood-Oxygen-Level-Dependent (BOLD) fMRI. After psilocybin administration, significant changes in consciousness were observed, accompanied by unexpected decreases in cerebral blood flow and BOLD signals, particularly in the thalamus and anterior/posterior cingulate cortex (ACC and PCC). A consistent finding was the reduced activity in the ACC and medial prefrontal cortex (mPFC), which correlated with the subjective intensity of the experience. Additionally, psilocybin significantly reduced positive connectivity between the mPFC and PCC, suggesting that the subjective effects of psilocybin arise from decreased activity and connectivity in key brain hubs, facilitating unconstrained cognition [113]. Complementary translational evidence comes from large-animal models. In Raval et al.'s study on pigs, a single intravenous dose of psilocybin (0.08 mg/kg) increased synaptic vesicle protein 2A (SV2A) density and reduced 5-HT2AR density. One day after treatment, SV2A rose in the hippocampus, while 5-HT2AR levels dropped sharply in both the hippocampus and prefrontal cortex (PFC). By day seven, SV2A remained elevated in both regions, but 5-HT2AR levels had returned to baseline. This suggests psilocybin induces lasting synaptogenesis alongside a transient reduction in 5-HT2AR density [114].

At the cellular level, Shao et al. used chronic two-photon microscopy to track dendritic spines of layer 5 pyramidal neurons in the mouse medial frontal cortex. A single psilocybin dose increased spine size and density by about 10%, driven by enhanced spine formation. These changes appeared within 24 h and persisted for at least a month. Psilocybin also improved stress-related behavioral deficits and boosted excitatory neurotransmission. The findings suggest that psilocybin rapidly and durably remodels cortical synapses, creating a structural basis for long-term experience integration and lasting therapeutic effects [62].

At the receptor level, psilocybin decreases hippocampal 5-HT1A receptor protein expression across both low and high doses, while producing dose-dependent bidirectional effects on 5-HT2A receptor density, with reductions observed at lower doses and increases at higher doses. In addition, psilocybin selectively upregulates GluN2A protein expression in the frontal cortex, without altering GluN2B or AMPA receptor subunits. In light of the central roles of NMDA, 5-HT1A, and 5-HT2A receptors in maintaining E/I balance, psilocybin-induced modulation of these receptor systems likely contributes to sustained reorganization of E/I dynamics following treatment [115]. Given that prenatal stress impairs both structural and functional neuroplasticity [116, 117], these converging findings suggest that psilocybin-induced plasticity may counteract, or partially compensate for, stress-related neurodevelopmental vulnerabilities, although this hypothesis remains to be directly tested.

Rebalancing E/I dynamics

A growing body of evidence suggests that psilocybin may directly influence the **E/I balance** that is often disrupted by prenatal stress and implicated in ADHD. Activation of 5-HT2A receptors on cortical pyramidal neurons increases glutamatergic transmission, but this effect is tightly regulated by local GABAergic interneurons, particularly parvalbumin-positive cells, leading to enhanced network synchrony [111, 118].

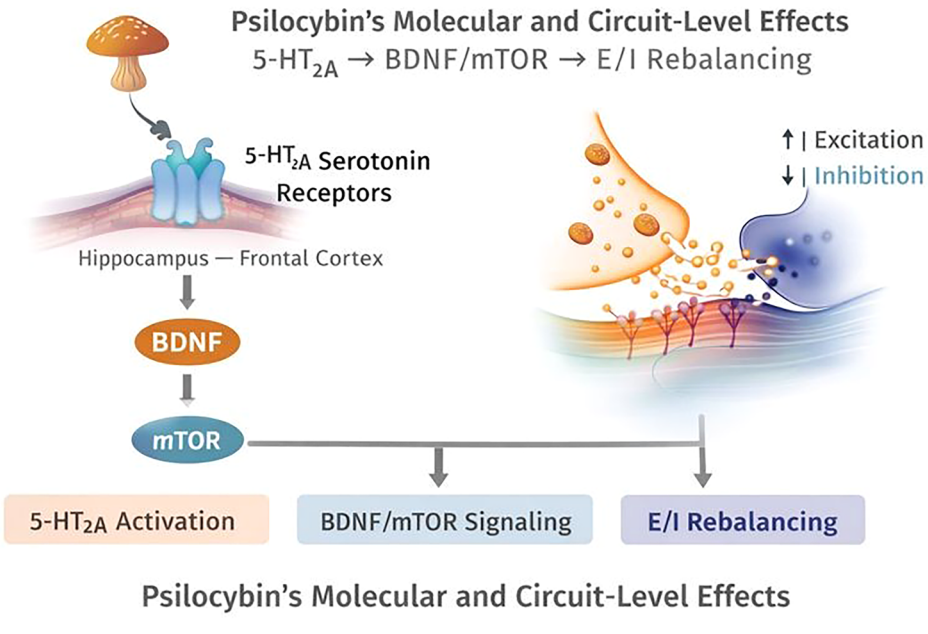


Fig. 2 Conceptual model of psilocybin-induced molecular and circuit-level plasticity. This schematic summarizes the proposed cascade linking psilocybin's primary molecular targets to circuit-level outcomes. Psilocybin activates cortical and hippocampal 5-HT₂A receptors, triggering downstream signaling pathways involving brain-derived neurotrophic factor (BDNF) and mammalian target of rapamycin (mTOR). Activation of these plasticity-related pathways is hypothesized to promote synaptic remodeling and restore excitatory–inhibitory (E/I) balance within stress- and cognition-relevant neural circuits, including frontal and hippocampal networks. The figure depicts a mechanistic framework rather than a confirmed causal pathway and is intended to guide future experimental investigation.

In animal models, psychedelics restore inhibitory tone by normalizing interneuron activity and increasing oscillatory coherence in prefrontal–hippocampal circuits [119]. These findings suggest that psilocybin may act as a **circuit-level modulator**, promoting flexible reorganization of dysfunctional networks while preventing runaway excitatory activity. Consistent with this, psilocybin, acting as a 5-HT2A receptor agonist on cortical pyramidal cells, has been shown to elevate extracellular glutamate levels in both the frontal cortex and hippocampus, which in turn stimulates GABAergic interneurons in the prefrontal cortex, leading to enhanced GABA release and restoration of cortical E/I balance [120]. This process helps restore the brain's E/I balance, especially within the cortical regions (Fig. 2). Supporting evidence from Wojtas et al. (2022) further demonstrated that a single dose of psilocybin increased extracellular dopamine, serotonin, glutamate, and GABA in the frontal cortex, while uniquely elevating GABA levels in the thalamic reticular nucleus. This coordinated modulation of excitatory and inhibitory systems mirrors the effects of ketamine but through distinct serotonergic mechanisms, underscoring psilocybin's capacity to stabilize network dynamics while preserving flexibility [121].

An important next step in refining the clinical relevance of psilocybin for ADHD is to link its network-level effects to specific symptom dimensions, including attentional deficits, impulsivity, and emotional dysregulation. ADHD pathophysiology has been associated with altered functional interactions among frontostriatal circuits supporting attention and cognitive control, the salience network anchored in the anterior cingulate cortex and insula, and networks supporting executive regulation, consistent with dimensional models of ADHD symptomatology [122]. Although psilocybin's large-scale neural effects have been documented—such as disruption and reconfiguration of default mode and other resting-state networks and modulation of connectivity patterns in regions implicated in cognition and emotion processing like the medial prefrontal cortex, anterior cingulate, and associated networks—there are no direct studies to date that link these network alterations to specific ADHD symptom dimensions [123]. Consequently, mechanistic inferences remain speculative. To address this gap, future studies could integrate symptom-specific behavioral phenotyping with multimodal imaging and electrophysiological assays in both preclinical models and clinical populations to determine whether variations in network-level responses covary with attentional control, impulsivity, and emotional dysregulation. This approach would help clarify whether psilocybin-induced modulation of specific circuits maps onto improvements in core ADHD symptom domains.

Mechanistic considerations and unresolved questions. Importantly, increases in both glutamatergic and GABAergic transmission raise critical questions regarding directionality and specificity of E/I correction. One possibility is that psilocybin enhances dynamic range and responsiveness of cortical circuits, allowing activity-dependent homeostatic mechanisms to recalibrate E/I balance regardless of whether baseline pathology reflects excess excitation or deficient inhibition. By strengthening inhibitory feedback and temporal precision, psilocybin may stabilize network activity even in the presence of elevated excitatory drive [115, 120]. However, it remains unclear whether this modulatory effect is universally corrective across all pathological states. In circuits where excitatory tone is already excessive or inhibitory control is severely compromised, psilocybin-induced glutamate release could, in principle, exacerbate dysregulation rather than restore balance. At present, there is insufficient experimental evidence to determine whether psilocybin can differentially normalize opposing forms of E/I imbalance or whether certain neurodevelopmental contexts may confer vulnerability to adverse effects. Accordingly, while existing data support psilocybin's capacity to influence E/I dynamics at the systems level, it remains unknown whether its effects are uniformly restorative or condition-dependent, particularly in prenatal stress–related or ADHD-

Table 1. Illustrative Evidence from Related Preclinical Animal Models: Psilocybin Effects on Neuroplasticity and E/I Balance (Not Including Prenatal Stress or ADHD Models).

References	Methods	Results	Comments
Shao et al. [62]	Single/brief psilocybin in mice; longitudinal 2-photon imaging; behavioral assays	Rapid formation and stabilization of dendritic spines; reversal of stress-induced social and structural deficits	Direct evidence of psilocybin-evoked structural plasticity relevant to stress pathology
Raval et al. [114]	Pigs; single 0.08 mg/kg psilocybin; in vitro autoradiography (SV2A)	↑ SV2A in hippocampus at 1 & 7 days	Large-mammal confirmation of synaptic plasticity marker increase
Jefsen et al. [143]	Rats; single psilocybin (0.5–20 mg/kg); RT-qPCR	PFC: ↑ cFos, FosB, Nr4a1, Psd95; Hippocampus: plasticity & IEG shifts	Acute, dose-dependent induction of plasticity-related genes
Wojtas et al. [121]	Rats; microdialysis (PFC, thalamus), comet assay, Western blot	↑ extracellular glutamate/5-HT; no genotoxicity; normalization of receptor proteins	Neurochemical facilitation without DNA damage; supports safety in brain tissue
Ly et al. [110]	Neuronal cultures & rodent models; psychedelics as 'psychoplastogens'	Robust neuritogenesis, spinogenesis, ↑ TrkB/mTOR signaling	Mechanistic foundation for psilocybin-like neuroplasticity
Hesselgrave et al. [124]	Rodent stress/depression models; behavioral + molecular endpoints	Antidepressant-like effects; plasticity-linked gene/protein modulation	Bridges stress phenotypes with psilocybin-induced plasticity
Cameron et al. [140]	Non-hallucinogenic psychoplastogen (TBG) in mice	Ketamine/psilocybin-like plasticity & behavioral rescue without hallucinations	Points to safer psilocybin-inspired analogs for developmental contexts
Peerboom & Ip [68]	Circuit-level analyses of plasticity/E–I mechanisms (preclinical)	Evidence that plasticity shifts restore E–I homeostasis	Mechanistic link to prenatal-stress–related E–I imbalance
Sohal & Rubenstein [34]	E–I balance frameworks; rodent circuit studies	E–I set-point restoration improves cognition	Provides rationale for psilocybin to target E–I in ADHD-relevant circuits
Kaila et al. [67]	GABAergic development & E–I maturation (preclinical)	Perturbed E–I during development → lasting dysfunction	Contextualizes prenatal stress vulnerability that psilocybin-evoked plasticity may correct

relevant pathophysiology. Targeted studies directly assessing baseline E/I states and their modulation by psilocybin will be essential to resolve these mechanistic uncertainties.

Preclinical and clinical evidence in stress-related conditions

Preclinical research indicates that psilocybin and related compounds can reverse stress-induced maladaptive plasticity. For example, chronic stress in rodents reduces dendritic complexity and GABAergic signaling in the prefrontal cortex, effects that are mitigated by psilocybin through increased synaptic density and improved behavioral outcomes [124]. Similarly, Ly et al. [110] demonstrated that a single administration of psilocybin promoted structural and functional plasticity in cortical neurons, enhancing dendritic spine growth and synaptogenesis, which are critical for stress resilience [110]. More recently, Shao et al. [62] reported that psilocybin normalized stress-induced changes in hippocampal neurogenesis and reduced anxiety-like behavior in mice, further supporting its role in reversing stress-related neuroadaptations [62]. Together, these cellular and circuit-level effects translate to behavioral resilience, including reduced anxiety-like and depressive behaviors.

In humans, psilocybin-assisted therapy has shown robust efficacy in treatment-resistant depression [125], end-of-life anxiety [126], post-traumatic stress disorder (PTSD), and substance use disorders[127]. Importantly, Carhart-Harris et al. [128] found that psilocybin facilitated sustained improvements in emotional regulation and social functioning in patients with treatment-resistant depression, effects associated with increased brain network flexibility [128]. Likewise, Mertens et al. [129] observed that psilocybin therapy significantly reduced stress-related rumination and improved well-being in patients with anxiety and depressive symptoms [129]. Although ADHD has not yet been a primary clinical target, the mechanisms implicated in psilocybin's therapeutic effects—enhanced neuroplasticity, improved emotional regulation, and network reorganization—are directly relevant to the cognitive and affective deficits observed in ADHD, particularly those linked to stress exposure and E/I imbalance.

Despite increasing interest in psilocybin's therapeutic potential, its clinical application requires careful consideration of safety and risk. Acute adverse effects, including transient anxiety, psychological distress, dissociation, and challenging subjective experiences, have been documented, particularly in vulnerable individuals [130]. Given that ADHD is characterized by impulsivity and emotional dysregulation [131], affected individuals may be at heightened risk for anxiety or behavioral disinhibition during psychedelic exposure. Accordingly, future clinical studies should incorporate thorough psychological screening, structured preparation, and post-session integration. Although available evidence indicates a low risk of physiological dependence and minimal abuse liability, tolerance development and appropriate re-dosing intervals remain important considerations, underscoring the need for rigorously controlled and ethically guided clinical protocols. To synthesize the most relevant evidence, we selected key studies addressing the relationship between psilocybin, neuroplasticity, and stress-related mechanisms, with particular consideration of their implications for prenatal stress and ADHD risk. Table 1 summarizes preclinical findings, highlighting animal models that demonstrate psilocybin's ability to restore E/I balance, enhance synaptic plasticity, and reverse stress-induced neurobiological and behavioral deficits. These studies provide a mechanistic foundation linking psilocybin to potential corrective effects in stress-sensitive neurodevelopmental pathways. Table 2 presents representative human research, including randomized controlled trials, neuroimaging studies, and translational reviews. Across these studies, psilocybin consistently promotes functional brain network reorganization, increases cognitive and emotional flexibility, and produces sustained therapeutic effects in stress-related psychiatric conditions. While direct evidence in prenatal stress or ADHD populations is not yet available, these converging preclinical and clinical data underscore psilocybin's promise as a candidate intervention.

Table 2. Illustrative Evidence from Related Clinical and Human Experimental Studies: Psilocybin Effects on Neuroplasticity and Stress-Related Mechanisms (Not Including ADHD Populations or Prenatal Stress Contexts).

References	Methods	Results	Comments
Griffiths et al. [126]	RCT in cancer-related anxiety/depression; psilocybin-assisted therapy	Large, durable reductions in anxiety/depression (months)	Clinical durability aligns with long-lasting plasticity
Carhart-Harris et al. [105]	TRD open-label; fMRI pre/post	Network 'reset', ↑ global connectivity post-psilocybin	Systems-level reorganization consistent with enhanced plasticity
Carhart-Harris et al. [119]	REBUS model + imaging synthesis	5-HT2A-driven network flexibility under psychedelics	Mechanistic scaffold linking serotonin, plasticity, and cognitive flexibility
Carhart-Harris et al. [113]	Healthy; psilocybin fMRI (rest/task)	DMN desegregation; altered coupling in limbic/association networks	Early imaging evidence of circuit-level reorganization
Davis et al. [125]	RCT in MDD; blinded comparison	Large effect sizes; rapid & sustained symptom reduction	Confirms robust clinical signal relevant to stress-linked pathology
Bogenschutz et al. [133]	Multisite RCT in alcohol use disorder	↓ Heavy drinking days vs control psychotherapy	Clinical change in stress-related compulsive pathology
Preller et al. [112]	Human psilocybin; dynamic functional connectivity	↑ Network flexibility/entropy; thalamocortical changes	Supports 'flexible brain' phenotype tied to plasticity
Mertens et al. [129]	Prospective/observational; psychological mediators	Emotional breakthrough & cognitive flexibility predict outcomes	Points to plasticity-mediated psychological change mechanisms
Vollenweider & Preller [109]	Translational review with imaging emphasis	Converges on 5-HT2A-mediated plasticity and network remodeling	Integrates preclinical–clinical plasticity narrative

Hypothesized preventive and corrective roles

Based on these mechanistic insights, psilocybin has been hypothesized to exert preventive or corrective effects on stress-related neurodevelopmental risk trajectories, including those associated with ADHD. By enhancing synaptic resilience and restoring inhibitory control, psilocybin could theoretically mitigate vulnerabilities arising from prenatal stress. Importantly, direct administration during pregnancy is neither ethical nor safe and is explicitly not proposed [132]. Instead, two speculative but conceptually grounded avenues are discussed:

1. **Preconception interventions:** Administering psilocybin-assisted therapy to prospective parents with histories of trauma, chronic stress, or psychiatric illness could reduce intergenerational transmission of stress vulnerability by improving emotional regulation, decreasing substance misuse, and fostering healthier parent–child interactions [133, 134].
2. **Postnatal interventions:** In offspring already exposed to prenatal stress, psilocybin could hypothetically enhance neuroplasticity during adolescence or adulthood, potentially mitigating attentional and emotional dysregulation before full ADHD symptomatology manifests [135, 136].

These hypotheses remain untested and should be interpreted as translational frameworks rather than clinical recommendations.

Ethical and developmental considerations

Despite growing scientific and clinical interest, psilocybin research in developmental contexts raises substantial ethical and neurodevelopmental concerns. Adolescence and early adulthood are critical periods of brain maturation marked by ongoing myelination, synaptic pruning, and large-scale network refinement [137]. Acute perturbation of serotonergic signaling and cortical excitability during these windows may carry unknown risks for long-term circuit development, cognitive maturation, and emotional regulation. At present, it remains unclear whether psilocybin-induced plasticity during these sensitive periods would be adaptive or potentially disruptive, warranting considerable caution. From a risk–benefit perspective, it is essential to recognize that established pharmacological treatments for ADHD, particularly stimulant medications, demonstrate response rates of approximately 70–80% and have well-characterized safety profiles [138]. Accordingly, experimental psychedelic-based approaches should not be viewed as alternatives to first-line therapies, but rather as speculative future options for treatment-resistant cases or for targeting neurobiological mechanisms insufficiently addressed by existing interventions. Research involving minors further raises complex ethical challenges related to informed consent and decisional capacity. Pediatric psychedelic studies would require careful distinction between adolescent assent and parental consent, alongside safeguards to mitigate coercion, therapeutic misconception, and undue influence [139]. The profound subjective effects of psilocybin may complicate informed decision-making, as minors may have limited capacity to anticipate or contextualize altered states of consciousness, highlighting the need for transparent risk communication and enhanced protective oversight. Regulatory barriers represent another major consideration. Psychedelics remain highly controlled substances in many jurisdictions, and pediatric trials would require extensive preclinical safety data, including developmental toxicology, long-term behavioral outcomes, and neurodevelopmental endpoints, before ethical approval could be considered. In this context, non-hallucinogenic psychoplastogens have been proposed as potentially safer alternatives capable of enhancing plasticity without inducing profound perceptual or psychological effects, although these compounds remain under active investigation [140]. Finally, ethical frameworks for vulnerable populations emphasize the principle of therapeutic sequencing. Lower-risk interventions—such as prenatal stress reduction strategies, early psychosocial and behavioral therapies, educational accommodations, and family-based interventions—should be prioritized and exhausted before considering experimental pharmacological approaches. Psychedelic research in developmental populations, if pursued at all, must therefore be incremental, evidence-driven, and firmly grounded in a hierarchy of risk minimization [141]. As psychedelics increasingly enter mainstream psychiatric discourse, ethical rigor must evolve in parallel with scientific innovation to ensure that emerging therapies are investigated responsibly, particularly

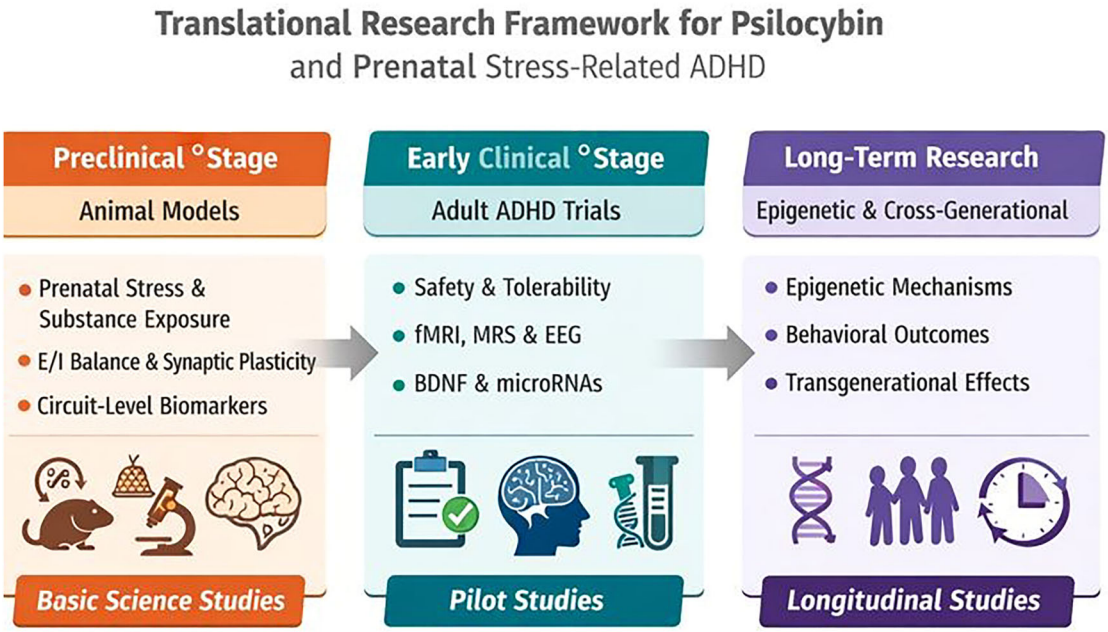


Fig. 3 Translational roadmap for investigating psilocybin in prenatal stress–related ADHD. This schematic illustrates a staged translational framework for future research on psilocybin in the context of stress-related and ADHD-relevant neurodevelopmental alterations. The preclinical stage emphasizes animal models of prenatal stress and substance exposure to identify circuit-level, synaptic, and excitatory–inhibitory (E/I) balance biomarkers. The early clinical stage focuses on carefully controlled pilot studies in adult ADHD populations, integrating safety assessments with multimodal biomarkers, including neuroimaging, electrophysiology, and molecular measures. The long-term research stage highlights longitudinal approaches aimed at understanding durable neuroplastic changes, including potential epigenetic and transgenerational mechanisms. Together, this framework underscores a stepwise, mechanistically informed, and ethically grounded pathway from basic science to clinical investigation.

when vulnerable populations and developing brains are involved [142].

CONCLUSION AND FUTURE DIRECTIONS

Accumulating preclinical and clinical evidence highlights the potential of psilocybin as a mechanistically informative and translationally relevant compound for stress-related neurodevelopmental and psychiatric conditions. Psilocybin's capacity to modulate E/I dynamics, enhance synaptic plasticity, and promote circuit-level reorganization positions it as a compelling candidate for addressing maladaptive neurobiological changes associated with prenatal stress—changes that are strongly implicated in ADHD and related disorders [104]. Rodent studies demonstrate that psilocybin can reverse stress-induced structural and functional deficits [62], while human trials consistently report robust and durable improvements in depression, anxiety, PTSD, and substance use disorders [127]. Although ADHD has not yet been the primary focus of psilocybin-based clinical trials, the core mechanisms underlying its therapeutic effects—enhanced neuroplasticity, improved emotional regulation, and increased network flexibility—map closely onto key cognitive and affective dimensions of ADHD pathology, particularly those linked to stress exposure and E/I imbalance. This convergence supports further investigation, but does not yet justify clinical application in ADHD populations. Despite its promise, substantial challenges must be addressed before psilocybin can be considered within developmental or preventative frameworks. Ethical and safety concerns are especially salient for vulnerable populations such as children and adolescents, where the long-term consequences of psychedelic-induced plasticity on brain maturation remain unknown. Moreover, the intensity and context-dependence of psychedelic experiences necessitate careful consideration of set, setting, therapeutic guidance, and post-session integration, alongside rigorous safety and monitoring protocols. In this regard, emerging research on non-hallucinogenic analogues (psychoplastogens) may offer a more developmentally appropriate strategy by preserving plasticity-enhancing effects while minimizing psychological risk [142].

Future research should prioritize mechanistic clarification rather than premature clinical translation. Preclinical models of prenatal psychological stress, substance exposure, and ADHD-like phenotypes provide a powerful platform to determine whether psilocybin can normalize E/I balance, enhance synaptic plasticity, and improve behavioral regulation within developmentally relevant circuits. In the short term, such models can be leveraged to define circuit-level, molecular, and epigenetic biomarkers of treatment response, integrating behavioral phenotyping with electrophysiology, receptor profiling, and stress-sensitive molecular markers. Longitudinal designs will be particularly critical to establish whether psilocybin-induced neuroplastic changes translate into sustained cognitive, emotional, and behavioral benefits across sensitive developmental windows, and whether these effects extend to intergenerational or epigenetic restoration mechanisms. From a clinical perspective, carefully controlled pilot studies in adult ADHD populations represent a logical and ethically appropriate next step. Mid-term clinical investigations should emphasize safety, tolerability, and target engagement, while integrating multimodal biomarkers—including functional and spectroscopic neuroimaging, electrophysiological measures, and peripheral or exosome-derived molecular indicators (e.g., BDNF, microRNAs)—to establish causal links between psilocybin-induced neurobiological changes and ADHD-relevant pathophysiology (Fig. 3). Such mechanistic validation will be essential before any consideration of younger or at-risk populations. Equally important is the proactive incorporation of ethical, developmental, and social considerations into study design, including informed consent, participant vulnerability, expectancy effects, and equitable access. Together, this staged, multimodal, and longitudinal framework provides a translational roadmap for advancing

psychedelic research in a scientifically rigorous and ethically responsible manner. Collaborative frameworks involving neuroscientists, clinicians, ethicists, educators, and families will be essential for shaping responsible and socially informed research trajectories.

In sum, psilocybin represents a powerful experimental tool in contemporary psychiatry, with particular relevance for stress-related—and potentially developmental—neuropsychiatric conditions. Its greatest near-term value may lie not as an immediate treatment for ADHD, but as a mechanistic probe for understanding how neuroplasticity, E/I balance, and stress-sensitive circuits can be modulated after vulnerability or dysfunction has emerged. Within this framework, psilocybin is best conceptualized as a candidate intervention for secondary and tertiary prevention, aimed at attenuating symptom progression, restoring cognitive and emotional regulation, and limiting long-term functional impairment rather than preventing initial disease onset. If pursued with scientific rigor, clinical restraint, and ethical responsibility, psilocybin and related compounds may ultimately open new avenues for developmentally informed interventions that target not only symptoms, but the underlying neurobiological mechanisms contributing to ADHD and related disorders.

REFERENCES

1. Mistry KB, Minkovitz CS, Riley AW, Johnson SB, Grason HA, Dubay LC, et al. A new framework for childhood health promotion: the role of policies and programs in building capacity and foundations of early childhood health. Am J Public Health. 2012;102:1688–96.
2. Torres-Soto NY, Corral-Verdugo V, Corral-Frías NS. The relationship between self-care, positive family environment, and human wellbeing. Wellbeing, Space Soc. 2022;3:100076.
3. Wade M, Wright L, Finegold KE. The effects of early life adversity on children's mental health and cognitive functioning. Transl Psychiatry. 2022;12:244.
4. National Academies of Sciences E, and Medicine Transforming the workforce for children birth through age 8: A unifying foundation. Washington, DC: The National Academies Press; 2015.
5. Luby JL, Rogers C, McLaughlin KA. Environmental conditions to promote healthy childhood brain/behavioral development: informing early preventive interventions for delivery in routine care. Biol Psychiatry Glob Open Sci. 2022;2:233–41.
6. Faye C, McGowan JC, Denny CA, David DJ. Neurobiological mechanisms of stress resilience and implications for the aged population. Curr Neuropharmacol. 2018;16:234–70.
7. Herzog JI, Schmahl C. Adverse childhood experiences and the consequences on neurobiological, psychosocial, and somatic conditions across the lifespan. Front Psychiatry. 2018;9:420.
8. Lewis E. Adverse childhood experiences and the developing brain. NCMH Retrieved. 2019;4:2023.
9. Swedo EAAM, Dahlberg LL, Niolon PH, Guinn AS, Simon TR, Mercy JA. Prevalence of adverse childhood experiences among U.S. adults—behavioral risk factor surveillance system, 2011–2020. Morbidity Mortal Wkly Rep. 2023;72:707–15.
10. DeSalvo KB, Volpp KG. Moving from a sickness model of health care to one of well-being—part 2. NEJM Catalyst. 2018;4. https://doi.org/10.1056/CAT.18.0148.
11. Benjamin RM. The national prevention strategy: shifting the nation's health-care system. Public Health Rep. 2011;126:774–76.
12. Barker DJ. In utero programming of chronic disease. Clin Sci. 1998;95:115–28.
13. Parrettini S, Caroli A, Torlone E. Nutrition and metabolic adaptations in physiological and complicated pregnancy: focus on obesity and gestational diabetes. Front Endocrinol. 2020;11:611929.
14. Vinnars M-T, Bixo M, Damdimopoulou P. Pregnancy-related maternal physiological adaptations and fetal chemical exposure. Mol Cell Endocrinol. 2023;578:112064.
15. Koyama T, Texada MJ, Halberg KA, Rewitz K. Metabolism and growth adaptation to environmental conditions in Drosophila. Cell Mol Life Sci. 2020;77:4523–51.
16. Karimi R, Yanovich A, Elbarbry F, Cleven A. Adaptive effects of endocrine hormones on metabolism of macronutrients during fasting and starvation: a scoping review. Metabolites. 2024;14:336.
17. Barker DJ, Forsén T, Uutela A, Osmond C, Eriksson JG. Size at birth and resilience to effects of poor living conditions in adult life: longitudinal study. Bmj. 2001;323:1273.
18. Etzel L, Apsley AT, Hastings WJ, Ye Q, Shalev I. Early life adversity is associated with differential gene expression in immune cells: a cluster-based analysis across an acute psychosocial stressor. Brain Behav Immun. 2024;119:724–33.
19. Gluckman PD, Hanson MA, Spencer HG. Predictive adaptive responses and human evolution. Trends Ecol evolution. 2005;20:527–33.
20. Dieckmann L, Czamara D. Epigenetics of prenatal stress in humans: the current research landscape. Clin Epigenetics. 2024;16:20.
21. Sandman CA, Davis EP, Buss C, Glynn LM. Exposure to prenatal psychobiological stress exerts programming influences on the mother and her fetus. Neuroendocrinology. 2012;95:8–21.
22. Saboory E, Mohammadi S, Dindarian S, Mohammadi H. Prenatal stress and elevated seizure susceptibility: Molecular inheritable changes. Epilepsy Behav. 2019;96:122–31.
23. O'Donnell KJ, Meaney MJ. Fetal origins of mental health: the developmental origins of health and disease hypothesis. Am J Psychiatry. 2017;174:319–28.
24. Shonkoff JP, Garner AS, Child CoPAo, Family Health CoEC, Adoption, Dependent Care, Developmental So, et al. The lifelong effects of early childhood adversity and toxic stress. Pediatrics. 2012;129:e232–e46.
25. Shonkoff JP, Boyce WT, McEwen BS. Neuroscience, molecular biology, and the childhood roots of health disparities: building a new framework for health promotion and disease prevention. Jama. 2009;301:2252–59.
26. Child HCotD. The Foundations of Lifelong Health Are Built in Early Childhood. Cambridge, MA: Harvard University; 2010.
27. Meaney MJ. Epigenetics and the biological definition of gene× environment interactions. Child Dev. 2010;81:41–79.
28. Knudsen EI. Sensitive periods in the development of the brain and behavior. J Cognit Neurosci. 2004;16:1412–25.
29. Baker AE, Galván A, Fuligni AJ. The connecting brain in context: how adolescent plasticity supports learning and development. Developmental Cognit Neurosci. 2025;71:101486.
30. Gee DG, Cohodes EM. Caregiving influences on development: a sensitive period for biological embedding of predictability and safety cues. Curr Dir Psychol Sci. 2021;30:376–83.
31. Blair C, Raver CC. Poverty, stress, and brain development: new directions for prevention and intervention. Acad Pediatrics. 2016;16:S30–6.
32. López-Vicente M, Szekely E, Lafaille-Magnan M-E, Morton JB, Oberlander TF, Greenwood CMT, et al. Examining the interaction between prenatal stress and polygenic risk for attention-deficit/hyperactivity disorder on brain growth in childhood: findings from the DREAM BIG consortium. Development Psychobiol. 2024;66:e22481.
33. Warm D, Schroer J, Sinning A. Gabaergic interneurons in early brain development: conducting and orchestrated by cortical network activity. Front Mol Neurosci. 2021;14:807969.
34. Sohal VS, Rubenstein JLR. Excitation-inhibition balance as a framework for investigating mechanisms in neuropsychiatric disorders. Mol Psychiatry. 2019;24:1248–57.
35. Lupien SJ, McEwen BS, Gunnar MR, Heim C. Effects of stress throughout the lifespan on the brain, behaviour and cognition. Nat Rev Neurosci. 2009;10:434–45.
36. Alotiby A. Immunology of stress: a review article. J Clin Med. 2024;13. https://doi.org/10.3390/jcm13216394.
37. Feng G, Xu X, Lei J. Tracking perceived stress, anxiety, and depression in daily life: a double-downward spiral process. Front Psychol. 2023;14:1114332.
38. Fruehwirth JCP KM, Ananat EO. Examining the effects of academic stress on student well-being in the transition to college. Human Soc Sci Commun. 2023;10:4698.
39. Association AP. Stress in America 2023: a nation recovering from collective trauma. American Psychological Association; washington, DC 2023.
40. McEwen BS, Akil H. Revisiting the stress concept: implications for affective disorders. J Neurosci. 2020;40:12–21.
41. Lupien SJ, Maheu F, Tu M, Fiocco A, Schramek TE. The effects of stress and stress hormones on human cognition: implications for the field of brain and cognition. Brain Cogn. 2007;65:209–37.
42. Koolhaas JM, Bartolomucci A, Buwalda B, de Boer SF, Flügge G, Korte SM, et al. Stress revisited: a critical evaluation of the stress concept. Neurosci Biobehav Rev. 2011;35:1291–301.
43. Karrow N. Activation of the hypothalamic–pituitary–adrenal axis and autonomic nervous system during inflammation and altered programming of the neuroendocrine–immune axis during fetal and neonatal development: lessons learned from the model inflammagen, lipopolysaccharide. Brain, Behav, Immun. 2006;20:144–58.
44. Van den Bergh BR, van den Heuvel MI, Lahti M, Braeken M, de Rooij SR, Entringer S, et al. Prenatal developmental origins of behavior and mental health: the influence of maternal stress in pregnancy. Neurosci Biobehav Rev. 2020;117:26–64.

45. Gheorghe CP, Goyal R, Mittal A, Longo LD. Gene expression in the placenta: maternal stress and epigenetic responses. Int J Dev Biol. 2010;54:507.
46. Graignic-Philippe R, Dayan J, Chokron S, Jacquet A, Tordjman S. Effects of prenatal stress on fetal and child development: a critical literature review. Neurosci Biobehav Rev. 2014;43:137–62.
47. Ayano G, Tesfaw G, Shumet S. Prevalence and determinants of antenatal depression in Ethiopia: a systematic review and meta-analysis. PLoS One. 2019;14:e0211764.
48. Wu Y, De Asis-Cruz J, Limperopoulos C. Brain structural and functional outcomes in the offspring of women experiencing psychological distress during pregnancy. Mol Psychiatry. 2024;29:2223–40.
49. Jagtap A, Jagtap B, Jagtap R, Lamture Y, Gomase K. Effects of prenatal stress on behavior, cognition, and psychopathology: a comprehensive review. Cureus. 2023;15:e47044.
50. Lund IO, Hannigan LJ, Ask H, Askelund AD, Hegemann L, Corfield EC, et al. Prenatal maternal stress: triangulating evidence for intrauterine exposure effects on birth and early childhood outcomes across multiple approaches. BMC Med. 2025;23:18.
51. Manzari N, Matvienko-Sikar K, Baldoni F, O'Keeffe GW, Khashan AS. Prenatal maternal stress and risk of neurodevelopmental disorders in the offspring: a systematic review and meta-analysis. Soc Psychiatry Psychiatr Epidemiol. 2019;54:1299–309.
52. Shao S, Wang J, Huang K, Wang S, Liu H, Wan S, et al. Prenatal pregnancy-related anxiety predicts boys' ADHD symptoms via placental C-reactive protein. Psychoneuroendocrinology. 2020;120:104797.
53. Ronald A, Pennell CE, Whitehouse AJ. Prenatal maternal stress associated with ADHD and autistic traits in early childhood. Front Psychol. 2011;1:223.
54. Ali AY, Inyang B, Koshy FS, George K, Poudel P, Chalasani R, et al. Elements that influence the development of attention deficit hyperactivity disorder (ADHD) in children. Cureus. 2022;14:e27835.
55. Association AP. Diagnostic and statistical manual of mental disorders: DSM-5-TR. 5th ed., text revision ed. Washington, DC: American Psychiatric Publishing; 2022.
56. Salari N, Ghasemi H, Abdoli N, Rahmani A, Shiri MH, Hashemian AH, et al. The global prevalence of ADHD in children and adolescents: a systematic review and meta-analysis. Italian J Pediatrics. 2023;49:48.
57. Faraone SV, Biederman J, Mick E. The age-dependent decline of attention deficit hyperactivity disorder: a meta-analysis of follow-up studies. Psychological Med. 2006;36:159–65.
58. van Lieshout M, Luman M, Twisk JW, van Ewijk H, Groenman AP, Thissen AJ, et al. A 6-year follow-up of a large European cohort of children with attention-deficit/hyperactivity disorder-combined subtype: outcomes in late adolescence and young adulthood. Eur Child Adolesc Psychiatry. 2016;25:1007–17.
59. Kessler RC, Adler L, Barkley R, Biederman J, Conners CK, Demler O, et al. The prevalence and correlates of adult ADHD in the United States: results from the National Comorbidity Survey Replication. Am J Psychiatry. 2006;163:716–23.
60. Caci HM, Morin AJ, Tran A. Prevalence and correlates of attention deficit hyperactivity disorder in adults from a French community sample. J Nerv Ment Dis. 2014;202:324–32.
61. Weibel S, Menard O, Ionita A, Boumendjel M, Cabelguen C, Kraemer C, et al. Practical considerations for the evaluation and management of Attention Deficit Hyperactivity Disorder (ADHD) in adults. L'encephale. 2020;46:30–40.
62. Shao LX, Liao C, Gregg I, Davoudian PA, Savalia NK, Delagarza K, et al. Psilocybin induces rapid and persistent growth of dendritic spines in frontal cortex in vivo. Neuron. 2021;109:2535–44.e4.
63. Moliner R, Girych M, Brunello CA, Kovaleva V, Biojone C, Enkavi G, et al. Psychedelics promote plasticity by directly binding to BDNF receptor TrkB. Nat Neurosci. 2023;26:1032–41.
64. Mueller-Buehl C, Wegrzyn D, Bauch J, Faissner A. Regulation of the E/I-balance by the neural matrisome. Front Mol Neurosci. 2023;16:1102334.
65. Ferguson BR, Gao WJ. PV interneurons: critical regulators of E/I balance for prefrontal cortex-dependent behavior and psychiatric disorders. Front Neural Circuits. 2018;12:37.
66. Ben-Ari Y, Khalilov I, Kahle KT, Cherubini E. The GABA excitatory/inhibitory shift in brain maturation and neurological disorders. Neuroscientist. 2012;18:467–86.
67. Kaila K, Ruusuvuori E, Seja P, Voipio J, Puskarjov M. GABA actions and ionic plasticity in epilepsy. Curr Opin Neurobiol. 2014;26:34–41.
68. Peerboom C, Wierenga CJ. The postnatal GABA shift: a developmental perspective. Neurosci Biobehav Rev. 2021;124:179–92.
69. Tang X, Jaenisch R, Sur M. The role of GABAergic signalling in neurodevelopmental disorders. Nat Rev Neurosci. 2021;22:290–307.
70. Giniatullin R, Cherubini E. Gabaergic signalling in Autism Spectrum disorders (ASD): role of glial cells and therapeutic perspectives. Brain Behav Immun. 2025;129:681–89.
71. Topchiy I, Mohbat J, Folorunso OO, Wang ZZ, Lazcano-Etchebarne C, Engin E. GABA system as the cause and effect in early development. Neurosci Biobehav Rev. 2024;161:105651.
72. Amin H, Marinaro F, De Pietri Tonelli D, Berdondini L. Developmental excitatory-to-inhibitory GABA-polarity switch is disrupted in 22q11.2 deletion syndrome: a potential target for clinical therapeutics. Sci Rep. 2017;7:15752.
73. Savardi A, Borgogno M, De Vivo M, Cancedda L. Pharmacological tools to target NKCC1 in brain disorders. Trends Pharmacol Sci. 2021;42:1009–34.
74. Hu L, Qiu M-J, Fan W-J, Wang W-E, Liu S-H, Liu X-Q, et al. Characterization of GABAergic marker expression in prefrontal cortex in dexamethasone induced depression/anxiety model. Front Endocrinol. 2024;15:1433026.
75. Hadler MD, Alle H, Geiger JRP. Parvalbumin interneuron cell-to-network plasticity: mechanisms and therapeutic avenues. Trends Pharmacol Sci. 2024;45:586–601.
76. Druga R, Salaj M, Al-Redouan A. Parvalbumin - positive neurons in the neocortex: a review. Physiol Res. 2023;72:S173–s91.
77. Korinek M, Candelas Serra M, Abdel Rahman F, Dobrovolski M, Kuchtiak V, Abramova V. et al. Disease-associated variants in GRIN1, GRIN2A and GRIN2B genes: insights into NMDA receptor structure, function, and pathophysiology. Physiol Res. 2024;73:S413–s34.
78. Marchisella F, Creutzberg KC, Begni V, Sanson A, Wearick-Silva LE, Tractenberg SG, et al. Exposure to prenatal stress is associated with an Excitatory/Inhibitory imbalance in rat prefrontal cortex and amygdala and an increased risk for emotional dysregulation. Front Cell Dev Biol. 2021;9:653384.
79. Jeon SC, Kim HJ, Ko EA, Jung SC. Prenatal exposure to high cortisol induces ADHD-like behaviors with delay in spatial cognitive functions during the post-weaning period in rats. Exp Neurobiol. 2021;30:87–100.
80. Roshan-Milani S, Seyyedabadi B, Saboory E, Parsamanesh N, Mehranfard N. Prenatal stress and increased susceptibility to anxiety-like behaviors: role of neuroinflammation and balance between GABAergic and glutamatergic transmission. Stress. 2021;24:481–95.
81. Rezaei Z, Jafari Z, Afrashteh N, Torabi R, Singh S, Kolb BE, et al. Prenatal stress dysregulates resting-state functional connectivity and sensory motifs. Neurobiol Stress. 2021;15:100345.
82. Mamiya PC, Richards TL, Edden RAE, Lee AKC, Stein MA, Kuhl PK. Reduced Glx and GABA inductions in the anterior cingulate cortex and caudate nucleus are related to impaired control of attention in attention-deficit/hyperactivity disorder. Int J Mol Sci. 2022;23:4677.
83. Puts NA, Ryan M, Oeltzschner G, Horska A, Edden RAE, Mahone EM. Reduced striatal GABA in unmedicated children with ADHD at 7T. Psychiatry Res Neuroimaging. 2020;301:111082.
84. Ferranti AS, Luessen DJ, Niswender CM. Novel pharmacological targets for GABAergic dysfunction in ADHD. Neuropharmacology. 2024;249:109897.
85. Yang MT, Fan HC, Lee HJ, Woudsma KJ, Lin KS, Liang JS, et al. Inter-subject gamma oscillation synchronization as a biomarker of abnormal processing of social interaction in ADHD. Sci Rep. 2024;14:17924.
86. Sylvester AL, Hensenne E, Ivanov D, Poser BA, Linden DEJ, van Amelsvoort T, et al. Neural excitation/inhibition imbalance and neurodevelopmental pathology in human copy number variant syndromes: a systematic review. J Neurodevelopmental Disord. 2025;17:31.
87. Mamiya PC, Arnett AB, Stein MA. Precision medicine care in ADHD: the case for neural excitation and inhibition. Brain Sci. 2021;11:91.
88. Coghill D, Banaschewski T, Cortese S, Asherson P, Brandeis D, Buitelaar J, et al. The management of ADHD in children and adolescents: bringing evidence to the clinic: perspective from the European ADHD Guidelines Group (EAGG). Eur Child Adolesc Psychiatry. 2023;32:1337–61.
89. Fortel I, Zhan L, Ajilore O, Wu Y, Mackin S, Leow A. Disrupted excitation-inhibition balance in cognitively normal individuals at risk of Alzheimer's disease. J Alzheimers Dis. 2023;95:1449–67.
90. Mehranpour A, Eskandari Z, Hejazi J, Gazori T, Saboory E. The effect of transcranial direct current stimulation (tDCS) and oxytocin on food craving and weight control in obese individuals with food addiction: a double-blind clinical trial. Int J Neurosci. 2025;136:189–201.
91. Mirzai N, Saboory E, Faridi A. Evaluating the effect of oxytocin and zinc sulphate on reproductive indices in opiate-addicted men receiving methadone maintenance therapy: a randomized, double-masked clinical trial. Addiction Health. 2025;17:1–7.
92. Estévez-Rodríguez M, Sánchez-Garrido Campos G, Zafra ÁM, Cordones I, Márquez-Ruiz J. Modulating excitation/inhibition balance through transcranial electrical stimulation: physiological mechanisms in animal models. Front Neurosci. 2025;19:1609679.
93. Fagerholm ED, Leech R, Williams S, Zarate CA Jr, Moran RJ, Gilbert JR. Fine-tuning neural excitation/inhibition for tailored ketamine use in treatment-resistant depression. Transl psychiatry. 2021;11:335.

94. Trakoshis S, Martínez-Cañada P, Rocchi F, Canella C, You W, Chakrabarti B, et al. Intrinsic excitation-inhibition imbalance affects medial prefrontal cortex differently in autistic men versus women. Elife. 2020;9:e55684.
95. Kessaris N, Magno L, Rubin AN, Oliveira MG. Genetic programs controlling cortical interneuron fate. Curr Opin Neurobiol. 2014;26:79–87.
96. Shoshani-Haye K, Friedlander G, Golani O, Almeida-Correa S, Shemesh Y, Chen A. Prenatal stress induces transient developmental alterations in distinct GABAergic populations and leads to long-lasting behavioral abnormalities. Neurobiol Stress. 2025;38:100749.
97. Marín O. Developmental timing and critical windows for the treatment of psychiatric disorders. Nat Med. 2016;22:1229–38.
98. Sadaghiani MM, Saboory E. Prenatal stress potentiates pilocarpine-induced epileptic behaviors in infant rats both time and sex dependently. Epilepsy Behav. 2010;18:166–70.
99. Mareckova K, Marecek R, Andryskova L, Brazdil M, Nikolova YS. Impact of prenatal stress on amygdala anatomy in young adulthood: timing and location matter. Biol Psychiatry Cog Neurosci Neuroimaging. 2022;7:231–38.
100. Hartman S, Belsky J, Pluess M. Prenatal programming of environmental sensitivity. Transl Psychiatry. 2023;13:161.
101. Bronson SL, Bale TL. The placenta as a mediator of stress effects on neurodevelopmental reprogramming. Neuropsychopharmacology. 2016;41:207–18.
102. Davis EP, Sandman CA. The timing of prenatal exposure to maternal cortisol and psychosocial stress is associated with human infant cognitive development. Child Dev. 2010;81:131–48.
103. Räikkönen K, Gissler M, Kajantie E. Associations between maternal antenatal corticosteroid treatment and mental and behavioral disorders in children. Jama. 2020;323:1924–33.
104. Calder AE, Hasler G. Towards an understanding of psychedelic-induced neuroplasticity. Neuropsychopharmacology. 2023;48:104–12.
105. Carhart-Harris RL, Goodwin GM. The therapeutic potential of psychedelic drugs: past, present, and future. Neuropsychopharmacology. 2017;42:2105–13.
106. Bonnieux JN, VanderZwaag B, Premji Z, Garcia-Romeu A, Garcia-Barrera MA. Psilocybin's effects on cognition and creativity: a scoping review. J Psychopharmacol. 2023;37:635–48.
107. Nichols DE, Johnson MW, Nichols CD. Psychedelics as medicines: an emerging new paradigm. Clin Pharmacol Ther. 2017;101:209–19.
108. Nutt D, Erritzoe D, Carhart-Harris R. Psychedelic Psychiatry's brave new world. Cell. 2020;181:24–28.
109. Vollenweider FX, Preller KH. Psychedelic drugs: neurobiology and potential for treatment of psychiatric disorders. Nat Rev Neurosci. 2020;21:611–24.
110. Ly C, Greb AC, Cameron LP, Wong JM, Barragan EV, Wilson PC, et al. Psychedelics promote structural and functional neural plasticity. Cell Rep. 2018;23:3170–82.
111. Inserra A, De Gregorio D, Gobbi G. Psychedelics in psychiatry: neuroplastic, immunomodulatory, and neurotransmitter mechanisms. Pharmacol Rev. 2021;73:202–77.
112. Preller KH, Duerler P, Burt JB, Ji JL, Adkinson B, Stämpfli P, et al. Psilocybin induces time-dependent changes in global functional connectivity. Biol Psychiatry. 2020;88:197–207.
113. Carhart-Harris RL, Erritzoe D, Williams T, Stone JM, Reed LJ, Colasanti A, et al. Neural correlates of the psychedelic state as determined by fMRI studies with psilocybin. Proc Natl Acad Sci USA 2012;109:2138–43.
114. Raval NR, Johansen A, Donovan LL, Ros NF, Ozenne B, Hansen HD, et al. A single dose of psilocybin increases synaptic density and decreases 5-HT(2A) receptor density in the Pig brain. Int J Mol Sci. 2021;22:835.
115. Kandilakis CL, Papatheodoropoulos C. Psychedelic modulation of excitation/inhibition balance: a dual-phase neurodevelopmental model. ACS Chem Neurosci. 2026;17:335–51.
116. Georgousopoulou V, Serdari A, Manomenidis G, Dacou-Voutetakis C. Prenatal stress, hormones, and fetal brain development: gender differences. Hormones. 2025.
117. Nolvi S, Merz EC, Kataja E-L, Parsons CE. Prenatal stress and the developing brain: postnatal environments promoting resilience. Biol Psychiatry. 2023;93:942–52.
118. Celada P, Puig MV, Artigas F. Serotonin modulation of cortical neurons and networks. Front Integr Neurosci. 2013;7:25.
119. Carhart-Harris RL, Friston KJ. REBUS and the anarchic brain: toward a unified model of the brain action of psychedelics. Pharmacol Rev. 2019;71:316–44.
120. Szpręgiel I, Bysiek A. Psilocybin and the glutamatergic pathway: implications for the treatment of neuropsychiatric diseases. Pharmacol Rep. 2024;76:1297–304.
121. Wojtas A, Bysiek A, Wawrzczak-Bargiela A, Szych Z, Majcher-Maślanka I, Herian M, et al. Effect of psilocybin and ketamine on brain neurotransmitters, glutamate receptors, DNA and Rat behavior. Int J Mol Sci. 2022;23:6713.
122. Hilger K, Fiebach CJ. ADHD symptoms are associated with the modular structure of intrinsic brain networks in a representative sample of healthy adults. Netw Neurosci. 2019;3:567–88.
123. Siegel JS, Subramanian S, Perry D, Kay BP, Gordon EM, Laumann TO, et al. Psilocybin desynchronizes the human brain. Nature. 2024;632:131–38.
124. Hesselgrave N, Troppoli TA, Wulff AB, Cole AB, Thompson SM. Harnessing psilocybin: antidepressant-like behavioral and synaptic actions of psilocybin are independent of 5-HT2R activation in mice. Proc Natl Acad Sci USA 2021;118:e2022489118.
125. Davis AK, Barrett FS, May DG, Cosimano MP, Sepeda ND, Johnson MW, et al. Effects of psilocybin-assisted therapy on major depressive disorder: a randomized clinical trial. JAMA Psychiatry. 2021;78:481–89.
126. Griffiths RR, Johnson MW, Carducci MA, Umbricht A, Richards WA, Richards BD, et al. Psilocybin produces substantial and sustained decreases in depression and anxiety in patients with life-threatening cancer: a randomized double-blind trial. J Psychopharmacol. 2016;30:1181–97.
127. Yaden DB, Griffiths RR. The subjective effects of psychedelics are necessary for their enduring therapeutic effects. ACS Pharmacol Transl Sci. 2021;4:568–72.
128. Carhart-Harris RL, Bolstridge M, Day CMJ, Rucker J, Watts R, Erritzoe DE, et al. Psilocybin with psychological support for treatment-resistant depression: six-month follow-up. Psychopharmacology. 2018;235:399–408.
129. Mertens LJ, Wall MB, Roseman L, Demetriou L, Nutt DJ, Carhart-Harris RL. Therapeutic mechanisms of psilocybin: changes in amygdala and prefrontal functional connectivity during emotional processing after psilocybin for treatment-resistant depression. J Psychopharmacol. 2020;34:167–80.
130. Studerus E, Kometer M, Hasler F, Vollenweider FX. Acute, subacute and long-term subjective effects of psilocybin in healthy humans: a pooled analysis of experimental studies. J Psychopharmacol. 2011;25:1434–52.
131. Shaw P, Stringaris A, Nigg J, Leibenluft E. Emotion dysregulation in attention deficit hyperactivity disorder. Am J Psychiatry. 2014;171:276–93.
132. Scott K, Lust K. Illicit substance use in pregnancy - a review. Obstet Med. 2010;3:94–100.
133. Bogenschutz MP, Ross S, Bhatt S, Baron T, Forcehimes AA, Laska E, et al. Percentage of heavy drinking days following psilocybin-assisted psychotherapy vs placebo in the treatment of adult patients with alcohol use disorder: a randomized clinical trial. JAMA Psychiatry. 2022;79:953–62.
134. Modlin NL, Creed M, Sarang M, Maggio C, Rucker JJ, Williamson V. Trauma-informed care in psychedelic therapy research: a qualitative literature review of evidence-based psychotherapy interventions in PTSD and psychedelic therapy across conditions. Neuropsychiatr Dis Treat. 2024;20:109–35.
135. Lepow L, Morishita H, Yehuda R. Critical period plasticity as a framework for psychedelic-assisted psychotherapy. Front Neurosci. 2021;15:710004.
136. Kuypers KP. Microdosing psychedelics in the treatment of ADHD and comorbid disorders. Eur Psychiatry. 2024;67:S7–S8.
137. Casey BJ, Tottenham N, Liston C, Durston S. Imaging the developing brain: what have we learned about cognitive development? Trends Cogn Sci. 2005;9:104–10.
138. Cortese S, Adamo N, Del Giovane C, Mohr-Jensen C, Hayes AJ, Carucci S, et al. Comparative efficacy and tolerability of medications for attention-deficit hyperactivity disorder in children, adolescents, and adults: a systematic review and network meta-analysis. Lancet Psychiatry. 2018;5:727–38.
139. Askari G, Vajdi M, Jafari-Nasab S, Golpour-Hamedani S. Ethical guidelines for human research on children and adolescents: a narrative review study. J Res Med Sci. 2024;29:53.
140. Cameron LP, Tombari RJ, Lu J, Pell AJ, Hurley ZQ, Ehinger Y, et al. A non-hallucinogenic psychedelic analogue with therapeutic potential. Nature. 2021;589:474–79.
141. Appelbaum PS. Clinical practice. Assessment of patients' competence to consent to treatment. N Engl J Med. 2007;357:1834–40.
142. Barber GS, Dike CC. Ethical and practical considerations for the use of psychedelics in psychiatry. Psychiatr Serv. 2023;74:838–46.
143. Jefsen OH, Elfving B, Wegener G, Müller HK. Transcriptional regulation in the rat prefrontal cortex and hippocampus after a single administration of psilocybin. J Psychopharmacol. 2021;35:483–93.

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SA and ES conceived and designed the study, conducted the literature search, collected the relevant articles for this review, and drafted the manuscript. All authors, including SR, contributed to writing the initial draft, critically revising the manuscript, and addressing the reviewers' comments. All authors read and approved the final version of the manuscript.

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ADDITIONAL INFORMATION

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